Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Spring-2019

Program: B.Sc. in CSE (Batch: 7^{th}) 1^{st} Year 2^{nd} Semester

Course Code: PHY 1201

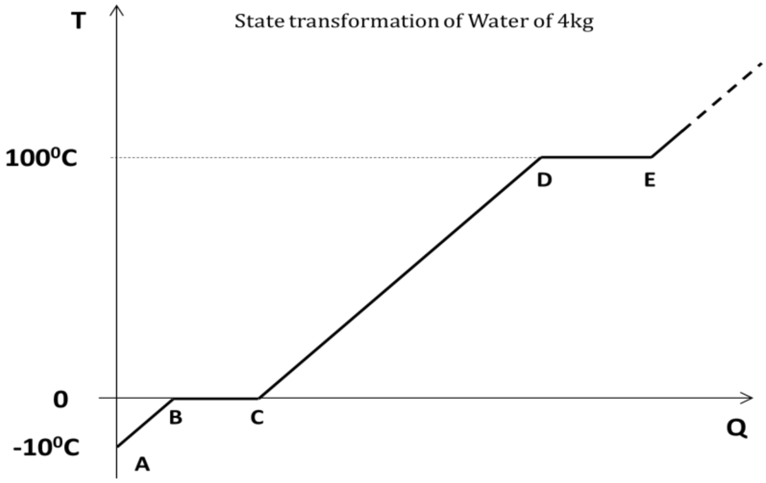
Course Title: Physics – II

Time: 1 hour & 30 minutes | Total Marks: 30

Group-B

[Answer any 5 (Five) questions including No.1] **5 × 5 = 25**

1. See the stem given below and answer the following questions:



a) Give the name of the state of A, B, C, D & E. 1

b) Find the change in entropy for the transformation from A to E. And also compare between BC & DE portion of above curve. 3+1=4

2. a) What may be the conditions to get an engine of 100% efficiency? 2

b) A heat engine after doing work in each cycle rejects 70% of heat absorbed from the source, calculate the efficiency of the engine. 3

3. a) “Adiabatic curve is more steeper than isothermal curve” - Explain it. 3

b) Calculate the values of molar specific heats of the ideal diatomic gases. 2

(R=8.31 $Jmol^{-1}K^{-1}$)

4. a) What are the basic differences between ideal gas equation & Van der Waal’s equation? 3

b) Establish the relation between temperature and pressure for adiabatic changes. 2

5. a) Why the platinum is the most commonly used material in resistance thermometer? 2

   b) A platinum resistance thermometer reads 2Ω and 2.73Ω at ice point and steam point respectively. Find the temperature at which it reads 4.83Ω. 3

6. a) What is Helmholtz free energy? 2

   b) Define enthalpy. 1

   c) Establish the relation between Helmholtz & Gibbs free energy. 2

Khwaja Yunus Ali University

School of Science and Engineering

Dept. of Computer Science and Engineering

Mid-term Examination: Spring-2019

Program: B.Sc. in CSE (Batch: 7th) 1st Year 2nd Semester

Course Code: PHY 1201

Course Title: Physics – II

Time: 1 hour & 30 minutes | Total Marks: 30

Group-A

[Answer all questions]

Time: 10 min | **0.5 × 10 = 5**

Student Name:	**ID No.:**

[Tick (√) mark the appropriate one]

1. According to the equipartition of energy, total energy of a rotational body will be .....

☐a) $\frac{1}{2}KT$ ☐b) $\frac{3}{2}KT$

☐c) $\frac{5}{2}KT$ ☐d) $\frac{7}{2}KT$

☐e) $\frac{9}{2}KT$

2. Most of the natural processes are .... Process.

☐a) reversible ☐b) irreversible

☐c) adiabatic ☐d) isothermal

☐e) isobaric

3. Efficiency of the Carnot's engine is ....

☐a) 10% ☐b) 40%

☐c) 50% ☐d) 70%

☐e) 100%

4. "Internal energy depends only on the temperature but not on pressure or volume" – stated by ....

☐a) Mayer ☐b) Joule

☐c) Van der Waal's ☐d) Sadi Carno

☐e) Brown

5. Which one is the correct relation between C_p & C_v?

☐a) $C_p > C_v$ ☐b) $C_p = C_v$

☐c) $C_p < C_v$ ☐d) $C_p + C_v = R$

☐e) $C_v - C_p = R$

6. What is the value of γ for Helium gas?

- ☐ a) 1.33
- ☐ b) 1.40
- ☐ c) 1.67
- ☐ d) 1.81
- ☐ e) 1.97

7. Mean free path is …………… the diameter of the molecule.

- ☐ a) independent to
- ☐ b) equal to
- ☐ c) proportional to
- ☐ d) inversely proportional to
- ☐ e) inversely proportional to square of

8. For isochoric process -

- ☐ a) dQ = 0
- ☐ b) dP = 0
- ☐ c) dS = 0
- ☐ d) dw = 0
- ☐ e) dU = 0

9. What is the unit of specific heat?

- ☐ a) $Jkg^{-1}K^{-1}$
- ☐ b) $calJ^{-1}$
- ☐ c) $Jcal^{-1}$
- ☐ d) JK^{-1}
- ☐ e) KJ^{-1}

10. How many steps included in the Carnot's cycle?

- ☐ a) 2
- ☐ b) 3
- ☐ c) 4
- ☐ d) 5
- ☐ e) 6

____

Invigilator Signature

____

Examiner Signature